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Semantic Shrinkage: Text-Similarity Priors for High-Dimensional Covariance Estimation

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Summary

This brief presents a method for estimating large covariance matrices using natural language processing. In a lookback ablation experiment spanning 75 quarterly rebalances (2007 to 2025), text-derived correlations combined with 252 days of sample volatilities produce a Sharpe ratio of 0.771, statistically significantly outperforming the Ledoit-Wolf estimator at the same lookback (block permutation p = 0.011) and exceeding Ledoit-Wolf with 504 days of full returns (Sharpe 0.540, p = 0.012). The method scales to universes of 2,000 or more assets where traditional approaches degrade, produces better-calibrated Value-at-Risk estimates, and requires no task-specific model training. Over 18 years that include the 2008 financial crisis, minimum-variance portfolios built with this text-based covariance beat the equal-weight baseline with roughly half the maximum drawdown (35% versus 57%). These findings replicate across three independent text sources: SEC 10-K filings, earnings call transcripts, and financial news articles.

1. The Problem and Our Approach

Portfolio construction requires estimating how asset returns move together, a task that reduces to estimating the covariance matrix. For a universe of p = 500 stocks, this matrix contains p(p-1)/2 = 124,750 unique pairwise entries, yet the estimation window may provide only 500 daily observations. When the number of assets approaches or exceeds the number of observations, sample covariance matrices become ill-conditioned, and portfolio optimizers amplify estimation errors into poor allocations.

The standard remedy, introduced by Ledoit and Wolf (2004), is shrinkage: blending the noisy sample covariance with a stable "target" matrix to regularize the estimate. The conventional target is a scaled identity matrix, which amounts to a prior that assumes all pairs of assets are uncorrelated. While this provides effective generic regularization, it discards any structural information about which assets should co-move.

Our approach replaces the identity target with a text-similarity matrix derived from corporate disclosures. We embed SEC 10-K annual filings, earnings call transcripts, and financial news articles using a pre-trained sentence transformer (BAAI/bge-base-en-v1.5, 768 dimensions; Xiao et al., 2024), then compute pairwise cosine similarity between firm embeddings. This similarity matrix serves as a structured prior: firms that describe themselves in similar language are expected to exhibit correlated returns. The covariance estimator thus blends the sample covariance with this semantic target, weighted by a shrinkage intensity parameter. We also examine a simpler variant, which we call *text_vols*, that uses the text-derived correlation matrix directly, combined with sample volatilities from a short return history. As we shall see in Section 4, this variant produces the strongest results and forms the basis of our headline findings.

No task-specific model was fine-tuned or trained for this work. The embedding model is frozen and used as-is, a deliberate choice that makes the method immediately deployable in any environment with access to corporate text. The pre-trained transformer captures linguistic patterns (for instance, that "cloud computing revenue" and "SaaS recurring revenue" describe related business activities) without any financial calibration.

The idea that textual similarity predicts return comovement has solid empirical foundations. Hoberg and Phillips (2010, 2016) demonstrated that firms with similar product descriptions in 10-K filings occupy neighboring positions in product market space, and their text-based industry classification has since become a standard tool in empirical finance. Dyer, Roulstone, and Van Buskirk (2024) showed more recently that disclosure similarity not only predicts but actively influences future return comovement, providing a causal mechanism for the relationship we exploit. The specific idea of using text similarity as a covariance shrinkage target was introduced by Becquin and Esmeir (2023), who demonstrated its feasibility at EMNLP using 10-K descriptions with the standard Ledoit-Wolf oracle intensity. Lu, Ndiaye, and Simaan (2024) pursued a related direction, substituting Term Frequency-Inverse Document Frequency (TF-IDF) similarity from the Hoberg-Phillips network and replacing the oracle with reinforcement learning for intensity selection. Separately, Gabaix, Koijen, Richmond, and Yogo (2025) constructed asset embeddings from portfolio holdings data rather than text, while Gawronsky and Huang (2024) modeled assets as distributions over embedded news using energy distance rather than cosine similarity.

Our work extends this line of research in several directions that, to our knowledge, have not been explored. We conduct the first systematic lookback ablation quantifying how much return history text can replace. We demonstrate scalability to 2,000 assets, where the semantic advantage doubles the Sharpe ratio of the standard estimator. We show that the Ledoit-Wolf oracle formula is substantially miscalibrated for informative targets, prescribing roughly 4% weight when the optimum lies between 25% and 75%, and resolve this with cross-validated intensity selection that significantly outperforms the standard estimator ($p = 0.034$). We provide the first VaR calibration tests of text-based covariance, and the first shuffle placebo analysis distinguishing structural from informational benefits. We show that a TF-IDF baseline produces identical results to the neural model, confirming the signal is textual structure rather than pretrained model knowledge. We test under institutional constraints (sector limits, volatility targets) where the text-based advantage strengthens. Finally, we replicate across three independent text sources (SEC filings, earnings transcripts, and financial news) whereas prior work relies on a single source. Nakayama, Sawaki, Furuya, and Tamura (2024) have begun extending text-based correlation to cross-asset allocation, a direction we discuss in Section 6.

2. Experimental Design

We now turn to the design of our experiments, beginning with the data and then describing the experimental protocols.

The investable universe consists of up to 500 of the most liquid US equities at each quarterly rebalance, selected from approximately 4,592 firms with 10-K filings. Eligibility requires that a firm's trailing 21-day median daily dollar volume exceed $1 million and that at least 80% of its daily returns be non-missing over the trailing 504-day lookback window. The evaluation period spans 18 years, from April 2007 to December 2025, comprising 75 quarterly rebalances. Three text sources provide independent signals: SEC 10-K annual filings (4,592 tickers, 60,258 documents), earnings call transcripts (608 tickers, 30,596 documents), and financial news articles (4,024 tickers, 111,062 quarterly aggregated documents).

Each experiment addresses a distinct research question. First, the main backtest ($p = 500$, 75 quarters) asks whether semantic shrinkage improves on Ledoit-Wolf in standard minimum-variance portfolio construction. Second, the lookback ablation ($p = 500$, 75 quarters, nine lookback windows ranging from 0 to 504 days) measures how much return history text can replace; this is the headline experiment, as it quantifies the information content of text in return-equivalent units. Third, a scaling experiment (p ranging from 50 to 2,000 over 75 quarters) tests whether the semantic advantage grows as traditional methods degrade at higher dimensionality. Fourth, a VaR calibration study (p from 200 to 2,000) evaluates whether the text-based covariance produces better-calibrated tail-risk estimates. Fifth, an alpha sweep ($p = 200$ and $p = 500$) maps the performance surface across shrinkage intensities to determine whether the standard Ledoit-Wolf formula is miscalibrated for informative targets. Sixth, cross-validated alpha calibration ($p = 500$, 71 rebalances) tests whether an adaptive, out-of-sample intensity selection can resolve the oracle miscalibration. Seventh, constrained optimization tests whether the semantic advantage holds under institutional constraints (sector concentration limits and volatility targets). Eighth, multi-target shrinkage ($p = 200$ and $p = 500$) blends text, Standard Industrial Classification (SIC) sector, identity, and sample targets to test whether combining multiple priors outperforms any single target. Ninth, a TF-IDF baseline replaces the neural encoder with a bag-of-words model to test whether the signal comes from the pretrained model's knowledge or from textual structure alone. Tenth, a temporal sensitivity analysis splits the backtest into pre-training, training-contemporaneous, and post-training sub-periods to test for look-ahead bias. Finally, a battery of robustness tests covers shuffle placebo analysis, Fama-French six-factor regression, sub-period splits, block size sensitivity, transaction cost sensitivity, SIC sector comparison, cold-start simulation, and more. All experiments are benchmarked against Ledoit-Wolf shrinkage (identity target), the equal-weight portfolio (1/N), SIC sector covariance, and the SPY exchange-traded fund.

3. Backtesting Rigor

A financial practitioner encountering these results will immediately ask how we know they are not overfit. This section addresses that concern directly. Every claim that follows was verified by an independent audit conducted by five specialized AI review agents covering data integrity, look-ahead bias, statistical methodology, code quality, and paper-code consistency.

The most fundamental safeguard is the walk-forward design. Every quarterly rebalance uses only data that is strictly prior to the rebalance date, enforced throughout all 15 experiment scripts via strict inequality filters. Embeddings use actual SEC EDGAR filing dates rather than fiscal year-end dates; consequently, a 10-K for fiscal year 2022 that was filed in March 2023 is not available to the estimator until after that filing date. The independent look-ahead bias audit rated this aspect of the codebase GREEN, indicating no violations were found.

Equally important, no hyperparameter was tuned on test data. All parameters were fixed before running any backtest, including the lookback window length, the maximum position weight, the block size for statistical testing, and the embedding model itself. The shrinkage intensity uses the Ledoit-Wolf oracle formula, applied identically to both the semantic and identity-target estimators. Because only the target matrix differs between the two methods, any performance difference is attributable solely to the quality of the prior, not to differential tuning.

The investable universe is determined once per quarter through ADV ranking, liquidity filtering, and completeness filtering, then shared across all strategies. Any survivorship bias therefore affects absolute return levels equally and does not contaminate relative comparisons. Transaction costs are stratified by liquidity tier (10 basis points for large-cap firms with average daily volume above $500 million, 20 basis points for mid-cap firms above $50 million, and 50 basis points for small-cap firms below that threshold), with a volatility regime multiplier capped at 2x that adjusts for periods of market stress. We tested sensitivity across cost multipliers from 0.5x through 2.0x and found the semantic advantage to be robust and slightly increasing at higher costs.

The statistical testing framework respects the autocorrelation structure of financial returns, which invalidates naive t-tests. Our primary test is a block sign-flip permutation test following Politis and Romano, using 63-day non-overlapping blocks and 10,000 permutations, which avoids the pitfall of treating dependent observations as independent. We complement this with Lo's (2002) Sharpe ratio correction for serial autocorrelation, Fama-French six-factor regressions with Newey-West heteroskedasticity and autocorrelation consistent standard errors, Benjamini-Hochberg false discovery rate corrections wherever multiple hypotheses are tested simultaneously, and Kupiec's (1995) unconditional coverage test for VaR model calibration.

The code audit yielded GREEN ratings for look-ahead bias, code quality, and paper-code consistency (with more than 200 numeric claims verified against source JSON files, achieving a 100% match rate). Statistical rigor received a YELLOW rating, primarily because some tests were underpowered with only 45 quarterly rebalances; this concern was addressed by extending key experiments to 75 quarters.

Several limitations should be noted. The pre-trained embedding model was trained on a text corpus that potentially includes material from future periods, which is standard practice in the NLP-finance literature. Two experiments address this concern definitively. A TF-IDF baseline, using only corpus-internal term frequencies with zero pretrained model knowledge and refitted at each rebalance, produces statistically indistinguishable results from the neural embeddings (Sharpe 0.592 versus 0.592, $p = 0.849$ over 75 quarters), confirming the signal arises from textual structure rather than model training data. A temporal sensitivity analysis corroborates this: the semantic advantage is significant in the pre-training period 2007-2016 ($p = 0.027$) and essentially zero in the training-contemporaneous period 2017-2022, the opposite of what look-ahead bias would produce. Tests at 45 quarters yield only approximately 11 effective independent blocks, motivating the extension to 75 quarters (approximately 17 blocks). The Ledoit-Wolf oracle formula is calibrated for identity targets rather than semantic targets, which makes our test conservative in the sense that the semantic method does not benefit from a custom-tuned intensity. The lookback ablation tests nine lookback windows, and the reported $p = 0.011$ at $L = 252$ has not been corrected for these multiple comparisons; while the structural finding that text_vols outperforms Ledoit-Wolf at all nine windows is itself informative, individual p-values should be interpreted cautiously. Finally, missing returns are zero-filled consistently across all strategies, with the 80% completeness filter limiting this to at most 20% of any individual series.

4. Findings

Having established the methodological foundations in the preceding sections, we now present the main results. The central finding concerns the decomposition of covariance information into correlation structure and volatility.

In the lookback ablation experiment (75 quarters, 2007 to 2025), the text_vols variant, which combines text-derived correlations with sample volatilities, achieves a Sharpe ratio of 0.771 when given $L = 252$ days of return data for volatility estimation. This result statistically significantly outperforms Ledoit-Wolf at the same lookback (block permutation $p = 0.011$) and exceeds Ledoit-Wolf with the full 504-day lookback (Sharpe 0.540, $p = 0.012$). Moreover, with only 63 days of volatility data, text_vols still achieves a Sharpe ratio of 0.713, which exceeds the Ledoit-Wolf estimate computed from two full years of returns. The result replicates with transcript embeddings, where text_vols peaks at $L = 63$ (Sharpe 0.735, surpassing equal weight at 0.624). It is important to note, however, that the main backtest comparison using semantic shrinkage with the standard oracle intensity shows a smaller gap (Sharpe 0.592 versus 0.540, $p = 0.239$). The headline result thus comes from text_vols, which uses the text prior more aggressively than the oracle formula prescribes.

The extended 75-quarter window also yields what appears to be the first instance of a covariance-optimized strategy outperforming equal weight in our experimental setup. Semantic shrinkage beats the 1/N baseline on Lo-corrected Sharpe (10-K: 0.716 versus 0.687; transcript: 0.906 versus 0.754), though we note that raw Sharpe margins are narrow (10-K: 0.592 versus 0.582, $p = 0.073$) and not individually significant. The drawdown advantage is more pronounced (35% versus 57%), but this reflects the benefit of minimum-variance construction broadly rather than the text prior specifically, since semantic and Ledoit-Wolf drawdowns differ by less than one percentage point. These observations are consistent with the reversal of the equal-weight dominance documented by DeMiguel, Garlappi, and Uppal (2009), which holds in benign markets but breaks down during crises where minimum-variance construction provides downside protection.

Turning to the question of scalability, at $p = 2{,}000$ assets (a p/n ratio of approximately 4.0), semantic shrinkage delivers nearly double the Sharpe ratio of Ledoit-Wolf (0.30 versus 0.15). The gap widens monotonically from +0.005 at $p = 50$ to +0.148 at $p = 2{,}000$. This pattern matters most for practitioners managing broad universes where estimation error is severe. In a closely related finding, the VaR calibration experiment demonstrates that semantic covariance is the best-calibrated method across all universe sizes and VaR levels. At 5% VaR with $p = 1{,}000$, the semantic violation ratio is 1.03 (nearly perfect calibration) compared to 1.24 for Ledoit-Wolf. At 2.5% VaR with $p = 2{,}000$, semantic achieves a violation ratio of 1.27 compared to 1.71 for Ledoit-Wolf. While all methods are rejected by the Kupiec unconditional coverage test at the 2.5% level after fence-post correction, semantic covariance is consistently closest to the nominal coverage rate. Standard covariance methods thus produce under-calibrated risk estimates at high dimensionality, whereas text-based priors provide better-conditioned matrices.

The alpha sweep experiment reveals that the Ledoit-Wolf oracle prescribes approximately 4% weight on the semantic target when the optimum lies between 25% and 75%. Over 75 quarters at $p = 500$, setting alpha to 0.25 achieves a Sharpe ratio of 0.752 ($p = 0.002$ versus Ledoit-Wolf), while pure text (alpha = 1.0) achieves 0.744 compared to Ledoit-Wolf's 0.540, an improvement of 38%. All fixed alpha values from 0.25 through 1.0 significantly outperform Ledoit-Wolf (p-values 0.002 to 0.014). This suggests that the standard formula, which was derived for identity targets, is substantially miscalibrated when applied to informative shrinkage targets. Cross-validated shrinkage intensity selection resolves this problem adaptively: at each rebalance, the trailing window is split temporally 70/30 and the alpha minimizing out-of-sample portfolio variance is selected. CV-alpha achieves a Sharpe ratio of 0.677, significantly outperforming Ledoit-Wolf ($p = 0.034$), and independently discovers a mean alpha of 0.76 (median 1.0), confirming without any manual tuning that the semantic target deserves far more weight than the oracle prescribes.

Under institutional constraints, the semantic advantage strengthens. At a 15% sector concentration limit, semantic shrinkage achieves Sharpe 0.599 versus equal weight's 0.570. Under combined sector and volatility constraints, semantic achieves 0.574 versus equal weight's 0.543. The maximum drawdown advantage persists across all constraint regimes (approximately 35% versus 57%), meaning any institutional mandate with a drawdown limit below 40% would eliminate equal weight as a feasible strategy.

A multi-target shrinkage estimator blending text, SIC sector, identity, and sample components reveals that cross-validated calibration allocates approximately 60% to identity (generic regularization), 33% to the sample, and only 7% combined to text and SIC. At $p = 200$, this multi-target estimator (Sharpe 0.692) is the best non-SIC method. Frobenius-loss calibration, by contrast, assigns approximately 100% to the sample at both p values, confirming the same pathology affecting the Ledoit-Wolf oracle: distance-based loss functions cannot assess the economic value of structured priors.

The lookback ablation result replicates across a third independent text source. News article embeddings (111,062 quarterly documents, 4,024 tickers) yield the strongest correlation signal (Spearman rho = 0.254 versus 0.104 for 10-K filings) and produce statistically significant text_vols versus Ledoit-Wolf differences at $L = 126$ ($p = 0.047$), $L = 252$ ($p = 0.033$), and $L = 504$ ($p = 0.018$). News text_vols peaks at $L = 504$ (Sharpe 0.731), beating equal weight (0.597) at every lookback length of 21 days or more.

Three caveats temper these findings and deserve explicit acknowledgment. First, SIC sector codes outperform text-based methods at most lookback lengths, which suggests that simpler industry structure captures much of the same signal with less estimation noise. Second, at $p = 500$, a shuffle placebo test indicates that the advantage is structural (arising from the spectral shape of the cosine matrix) rather than firm-specific. At $p = 2{,}000$, however, this pattern reverses: 80% of shuffled runs underperform, suggesting that firm-specific text content matters more at higher dimensionality. Third, no strategy generates significant alpha after Fama-French six-factor adjustment, and the minimum-variance strategies tilt toward conservative, profitable firms (with a CMA loading of approximately 0.22). The performance differences we observe are therefore better understood as differences in covariance estimation quality rather than as evidence of novel return predictability.

5. Practical Implications

The findings described in Section 4 carry several practical implications for portfolio construction and risk management.

The most immediate consequence concerns data requirements. New assets, initial public offerings (IPOs), and post-restructuring entities have text (prospectuses, filings) available immediately, whereas return histories take years to accumulate. As the lookback ablation demonstrates, text-based correlations with as few as 63 days of volatility data outperform the industry standard using 504 days of full returns. Even at $L = 0$, meaning zero return history, text-only covariance still produces viable allocations with a Sharpe ratio of 0.568. This capability is particularly relevant for portfolio managers who must incorporate newly listed securities, newly covered firms, or instruments emerging from corporate restructuring.

The VaR and scaling advantages translate directly to regulatory and institutional needs. For institutions computing Basel III regulatory VaR on large portfolios, text-based covariance provides the well-conditioned matrices that sample-based methods cannot deliver at high dimensionality. Recall from Section 4 that semantic covariance achieves the lowest VaR violation ratios across all universe sizes, with near-perfect calibration at the 5% level (violation ratio 1.03 at $p = 1{,}000$). The method scales to 2,000 or more assets, making it applicable to large institutional portfolios, multi-strategy funds, and index providers.

From an implementation standpoint, the method is straightforward to deploy. It requires off-the-shelf sentence embeddings (with no task-specific training), or even a simple TF-IDF pipeline (which produces equivalent results), quarterly rebalancing, and standard quadratic programming via CVXPY with the OSQP solver. It integrates into existing portfolio infrastructure as a drop-in replacement for the Ledoit-Wolf shrinkage target, sharing the same optimizer, constraints, and transaction cost model. The cross-validated alpha calibration removes the need for manual parameter selection: it discovers the appropriate text weight adaptively at each rebalance. Under realistic institutional constraints (sector limits, volatility targets), the text-based covariance advantage strengthens rather than weakens, and the drawdown protection persists across all constraint regimes.

We emphasize, however, that this approach is a complement to, not a replacement for, return-based estimation. The greatest value appears when return history is short, the universe is large (p/n exceeding 1), or during regime changes when historical correlations become unreliable. In such settings, text provides a stable structural prior that anchors the covariance estimate when return data is noisy or scarce.

6. Research Roadmap

We conclude by outlining directions for future work, beginning with the most promising extension and proceeding to more speculative possibilities.

The primary next direction is cross-asset covariance estimation using fused embedding spaces. The current work uses text exclusively for equities, but the data requirements problem is far more acute in other asset classes. Fixed income instruments (bond prospectuses, credit reports), commodities (supply reports, geopolitical text), and derivatives all have associated text but often lack synchronized return histories. A contrastive learning architecture, following the CLIP paradigm, could align text and return-series encoders into a shared embedding space, enabling cross-asset covariance estimation even between instruments that have never traded simultaneously. Potential extensions include climate and ESG text for green bond covariance and sovereign debt language for country risk matrices. We note, however, that no preliminary evidence yet supports these cross-asset applications; they remain conjectural.

A second direction concerns further refinement of shrinkage calibration. Cross-validated alpha selection already significantly outperforms the Ledoit-Wolf oracle ($p = 0.034$), but Bayesian approaches or analytical corrections specifically derived for informative targets could improve further while reducing computational overhead.

Third, while the TF-IDF equivalence result strongly suggests that temporal model retraining would not change the findings, training the language model at each rebalance using only text available up to that point would constitute the gold-standard elimination of any remaining concern about anachronistic model knowledge. This approach is computationally expensive but would close the question definitively.

Finally, two operational extensions merit exploration: higher-frequency rebalancing triggered by news and transcript release dates, which could capture regime shifts faster than calendar-based quarterly scheduling, and live deployment through paper trading with real data feeds, actual execution costs, and operational constraints.

All numeric claims in this brief trace to versioned JSON result files in the project repository. The full methodology, all 27 experiments (including 19 robustness tests), and complete reproduction instructions are available in the accompanying research paper.

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